MetaGaze: A Transformer-Based Multi-Feature Fusion for Robust Gaze Estimation

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Session 2022-2026

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June, 2026

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Acknowledgements

We would like to extend our heartfelt gratitude to everyone who played a part in the successful completion of this project.
Our deepest appreciation goes to our advisor, Ms. Sara Rehmat, with her constant guidance, insightful feedback, and encouragement were instrumental in shaping the direction of our work. Her dedication and expertise inspired us to strive for excellence throughout this journey.

We are also sincerely thankful to FAST for providing the resources, facilities, and academic environment that enabled us to carry out this project effectively. Our special thanks go to Mr. Riaz Nawab for his valuable assistance, collaboration, and willingness to share his time and knowledge whenever needed. We thank you all once again for your contributions and support.

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Abstract

Gaze estimation enables machines to understand where a person is looking, serving as a fundamental component for advanced human–computer interaction, driver monitoring, and assistive technologies. Despite notable advancements in this field, achieving highly accurate and reliable gaze prediction in natural, unconstrained environments remains a challenge due to variations in head pose, lighting conditions, and occlusions caused by glasses, shadows, or motion blur.

To overcome these limitations, this project introduces **MetaGaze**, a transformer-based framework designed to achieve robust and precise gaze estimation through multi-feature fusion. The model integrates facial, left-eye, and right-eye image features along with geometric head orientation vectors to capture both fine-grained and holistic spatial information. A symmetric cross-attention mechanism is employed to effectively fuse these complementary features, enabling the system to adapt seamlessly across diverse environmental and viewing conditions.

MetaGaze is trained and evaluated on a widely recognized benchmark dataset, ETH-XGaze, demonstrating strong generalization capability and superior estimation accuracy compared to existing state-of-the-art methods. Additionally, the system is optimized for real-time, camera-based gaze tracking, making it ideal for hands-free user interaction, foveated rendering, and other gaze-driven applications. By uniting efficiency, adaptability, and precision, MetaGaze offers a reliable and scalable solution for the next generation of gaze-controlled systems and intelligent user interfaces.

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CHAPTER 1

INTRODUCTION

1.1 Overview

Gaze direction represents the orientation of a person's visual focus in three-dimensional space, determined by the combined influence of eye movement, head orientation, and point of fixation. Accurate gaze estimation provides valuable insight into user attention, cognitive intent, and behavioral patterns, forming a key foundation for applications in human–computer interaction, healthcare, automotive safety, and immersive environments such as virtual and augmented reality.

1.1.1 Motivation

Although gaze estimation has broad and impactful applications, existing systems often struggle to maintain accuracy and stability in unconstrained, real-world settings. Variations in head movement, lighting conditions, and occlusions frequently disrupt prediction performance. The motivation behind this research is to develop a deep learning–based solution that can achieve robust and accurate gaze estimation using only a single RGB camera, eliminating the need for specialized or expensive hardware.

1.1.2 Background

Traditional gaze estimation methods primarily depended on geometric modeling and infrared-based hardware to measure eye movements with precision. While effective in controlled environments, these systems were often expensive, rigid, and impractical for everyday use. The rise of deep learning transformed the field by introducing appearance-based methods that rely on Convolutional Neural Networks (CNNs) to automatically learn visual features from facial and eye images. This shift made gaze tracking more accessible and flexible. However, CNNs struggle to capture long-range dependencies how different parts of an image relate to each other which limits their performance in dynamic, real-world settings. To overcome this, researchers have turned to transformer based architectures that use self-attention mechanisms to understand global context, enabling more accurate and adaptable gaze estimation.

1.1.3 Problem Statement

While deep learning based gaze estimation models have achieved remarkable progress, several challenges persist:

- Limited robustness to variations in head pose and lighting conditions.
- Dependence on clear, unobstructed eye images.
- Inconsistent performance and poor generalization across different datasets and environments.

These challenges hinder the deployment of gaze estimation in real-world, interactive, and accessibility-oriented systems.

1.1.4 Solution Overview

To address these limitations, this study introduces **MetaGaze** a transformer-based hybrid framework designed for accurate and reliable gaze estimation under natural conditions. MetaGaze employs a multi-feature fusion approach that integrates information

from facial, left-eye, and right-eye images, along with geometric head orientation vectors. Through a symmetric cross-attention mechanism, the model effectively combines local and global spatial cues, allowing it to adapt to diverse viewing conditions while maintaining high estimation accuracy.

MetaGaze aims to deliver a scalable, real-time solution suitable for gaze-controlled interfaces, hands-free applications, and next-generation intelligent systems that seamlessly bridge human attention with machine perception.

CHAPTER 2

REVIEW OF LITERATURE

Accurate gaze estimation forms the foundation for intelligent interaction systems, driver monitoring, and assistive technologies. Early approaches relied on geometric calibration, which performed well in controlled setups but failed in natural, unconstrained environments. With the advancement of deep learning, gaze prediction transitioned from handcrafted feature engineering to appearance-based models using CNNs, transformers, and hybrid fusion strategies. Convolutional Neural Networks (CNNs) enabled end-to-end learning of gaze features directly from image data. Lian et al. [1] proposed a multitask CNN for multiview gaze prediction using shared layers for direction and point estimation. Cheng et al. [2] extended this with FARE-Net, which modeled left-right eye asymmetry for improved spatial accuracy. Akinyelu and Blignaut [3] developed a lightweight CNN suitable for mobile devices, while Luo et al. [4] introduced CI-Net with cooperative feature fusion to mitigate partial occlusion issues. These CNN-based approaches achieved strong performance under controlled lighting but lacked generalization when subject or environment varied.

Transformers introduced attention mechanisms capable of capturing global relationships within visual data, marking a significant shift in the field. Cheng and Lu [5] presented GazeTR-Pure, the first transformer-only architecture for gaze estimation, using self-attention over image patches. Ye et al. [6] later proposed TransGaze, a Vision Transformer (ViT) that utilized pretrained weights for faster convergence and improved accuracy. Although

these models captured long-range dependencies effectively, they required substantial data and computational resources to achieve consistent generalization.

Recognizing these limitations, recent research demonstrates that combining CNNs for local feature extraction with transformers for global reasoning leads to superior results. Zhou et al. [7] proposed EM-Gaze, a two-stream CNN architecture integrating correlation and metric learning modules. Ren et al. [8] introduced MIEF, which fused multi-level CNN features but struggled under uneven lighting. Li et al. [9] designed Res-Swin-GE, a hybrid model combining ResNet and Swin Transformer backbones for improved cross-pose robustness. These studies collectively validate hybrid fusion as a promising pathway for robust and scalable gaze estimation, directly inspiring the architecture of **MetaGaze**.

2.1 Summary of Reviewed Studies

Table 2.1: Summary of representative gaze estimation studies

Year	Authors	Architecture / Model	Methodology	Limitations
2019	Lian et al. [1]	Multiview CNN	Multitask CNN jointly predicts gaze direction and point using multiview images.	Requires multiple camera inputs; lacks single-view adaptability.
2020	Cheng et al. [2]	FARE-Net (CNN)	Introduced asymmetric regression between eyes to improve 3D gaze accuracy.	Accuracy drops with occluded eyes or inconsistent lighting.
2022	Akinyelu and Blignaut [3]	Lightweight CNN	Developed efficient CNN optimized for mobile deployment.	Reduced precision for non-frontal head poses and illumination variation.
2022	Luo et al. [4]	CI-Net (Cooperative CNN)	Designed cooperative network combining consistent and inconsistent features for occlusion handling.	Limited robustness under severe occlusion and large head rotations.
2022	Cheng and Lu [5]	GazeTR-Pure (Transformer)	Utilized patch embedding and self-attention for transformer-based gaze regression.	Heavy computational load; requires extensive pretraining.
2023	Zhou et al. [7]	EM-Gaze (CNN Fusion)	Two-stream CNN combining eye and face correlation with metric learning.	Degraded performance under motion blur or extreme head tilt.
2023	Ren et al. [8]	MIEF (CNN Fusion)	Multi-level information fusion network improving gaze representation.	Sensitive to lighting variations; error accumulation in complex scenes.
2023	Li et al. [9]	Res-Swin-GE (Hybrid)	Combined CNN (ResNet) and Swin Transformer backbones for cross-pose estimation.	Moderate computational cost; complex integration pipeline.
2024	Ye et al. [6]	TransGaze (Transformer)	Vision Transformer leveraging pretrained weights for better generalization.	Large parameter count; less efficient on small datasets.

2.2 Connection to Proposed Work (MetaGaze)

From the reviewed literature, it is clear that CNNs extract detailed spatial features effectively, while transformers capture global context through attention mechanisms. However, neither approach alone achieves high robustness under real-world conditions. The proposed **MetaGaze** framework builds upon these insights by unifying facial, eye, and geometric head-orientation cues through symmetric cross-attention fusion, offering improved generalization and stability for real-time gaze estimation tasks.

CHAPTER 3

PROJECT VISION

3.1 Problem Statement

Today's gaze estimation systems often struggle when faced with the unpredictability of real-world settings. Rapid changes in head pose, uneven lighting, or partial occlusions can cause significant drops in accuracy. Most existing models depend heavily on either isolated eye images or full-face inputs, which restricts their ability to adapt across diverse scenarios. Moreover, deep learning approaches tend to favor one strength CNNs excel at fine-grained local details, while transformers capture broader contextual patterns but rarely balance both effectively.

These gaps make it difficult to deploy reliable gaze tracking in critical applications like driver safety systems, assistive technologies, or interactive interfaces. What's needed is a unified, efficient framework that intelligently combines multiple visual cues without demanding excessive computational resources.

3.2 Business Opportunity

MetaGaze addresses this need with a practical, software-only solution that works using ordinary RGB cameras—no infrared sensors, no multi-camera rigs. This accessibility unlocks value across industries: In healthcare, it can support non-invasive cognitive screening and patient attention monitoring. In automotive environments, it enables

smarter driver awareness systems. For accessibility, it empowers users with limited mobility through intuitive, gaze-driven controls.

By delivering high accuracy, real-time responsiveness, and cross-device compatibility, MetaGaze offers a versatile platform for both commercial products and academic exploration in human–computer interaction.

3.3 Objectives

This project aims to build a resilient, transformer-augmented system for precise gaze estimation through multi-source feature integration. The core goals include:

- Designing a hybrid model that seamlessly blends convolutional layers for local precision with transformer modules for global awareness.
- Fusing features from the face, both eyes, and head orientation using symmetric cross-attention to model interdependencies.
- Training and validating performance on challenging benchmark dataset like ETH-XGaze to ensure robust generalization.
- Deploying a live prototype that demonstrates smooth, hands-free interaction via a standard webcam.

3.4 Project Scope

The current phase focuses on designing the model architecture, training it effectively, and evaluating its performance using both publicly available and privately accessed datasets.The resulting system will support real-time inference on consumer hardware through an accessible webcam interface. Future work could expand to VR/AR integration, foveated rendering optimization, or embedded deployment in assistive devices.

Key deliverables include data preprocessing pipelines, the fusion-based network, comprehensive performance analysis, and an interactive demo application.

3.5 Constraints

Although access to the high-quality ETH-XGaze dataset significantly strengthens training potential, several practical challenges remain in developing and deploying **MetaGaze**:

- **Dataset Scale and Preprocessing:** While ETH-XGaze provides over one million diverse samples with extreme head poses, varied lighting, and natural occlusions, its sheer size demands efficient data pipelines, normalization strategies, and careful train/validation splits to avoid overfitting and ensure fair evaluation.
- **Hardware Efficiency:** The final system must deliver real-time performance on everyday devices like laptops or webcams, requiring aggressive model optimization despite the richness of training data.
- **Training Demands:** Large transformer modules still require extended training time and GPU memory, even with a strong dataset. Techniques such as gradient checkpointing, mixed precision, or staged training will be essential for feasible convergence.
- **Input Assumptions:** The system assumes consistent camera distance, stable user positioning, and minimal motion blur aligned with ETH-XGaze collection protocol.

3.6 Stakeholders Description

MetaGaze engages a range of beneficiaries and contributors:

- **End Users:** People with motor impairments or those seeking touchless control gain independence through natural eye-based navigation.
- **Healthcare Providers:** Doctors and therapists can leverage gaze patterns for early detection of attention deficits or neurological changes.
- **Automotive Industry:** Car manufacturers integrate reliable distraction detection to enhance road safety.

- **AR/VR Developers:** Immersive experiences improve via dynamic rendering focused on the user's point of regard.
- **Academic Researchers:** The open framework supports experimentation in attention modeling, multi-modal fusion, and efficient vision architectures.

3.6.1 Stakeholders Summary

End Users experience seamless, empowering control without physical input. **Healthcare professionals** access subtle behavioral insights through gaze analytics. **Automotive manufacturers** build safer vehicles with proactive monitoring. **AR/VR developers** create responsive, user-centered virtual worlds. **Researchers** gain a flexible, reproducible tool to push forward hybrid vision systems.

CHAPTER 4

SOFTWARE REQUIREMENTS SPECIFICATION

4.1 Key Features

MetaGaze is a research framework for gaze estimation from RGB images. It uses CNNs to extract features from eyes and face, and fuses all inputs with a transformer to predict gaze direction.

Key capabilities:

- Processes left eye, right eye, and full face regions with head orientation.
- Fuses features with cross-attention in a transformer.
- Outputs continuous gaze vectors.
- Built for ETH-XGaze (1M+ images with extreme poses and real-world variations).

4.2 Functional Requirements

4.2.1 Input Processing

Loads RGB images, detects face and eyes, computes head pose .

4.2.2 Feature Extraction

CNN encoders generate feature vectors for left eye, right eye, and full face.

4.2.3 Feature Fusion

Transformer integrates eye, face, and head pose features using cross-attention.

4.2.4 Gaze Prediction

Regression head outputs normalized gaze direction.

4.2.5 Evaluation

Computes angular error and per-subject performance on benchmark datasets.

4.3 Non-Functional Requirements

The following goals guide the development of **MetaGaze** as a reliable research contribution:

- **Accuracy:** Achieves a mean angular error (MAE) of approximately 3° on the ETH-XGaze test set.
- **Robustness to Head Pose:** Designed to handle large head rotations (yaw, pitch, roll) as seen in ETH-XGaze.
- **Robustness to Lighting:** Aims to maintain performance under different indoor lighting conditions (bright, dim, mixed).
- **Robustness to Occlusion:** Seeks to perform well when eyes are partially covered .
- **Reproducibility:** Experiments can be rerun with identical results using fixed random seeds.
- **Training Feasibility:** Model can be trained on ETH-XGaze using standard academic hardware (single GPU).

Table 4.1: Non-Functional Research Goals

Goal	**Description**
Accuracy	3° mean angular error (MAE) on ETH-XGaze test set
Head Pose Robustness	Effective across extreme head orientations in dataset
Lighting Robustness	Consistent performance under varied indoor illumination
Occlusion Handling	Graceful degradation with partial eye coverage
Reproducibility	Same results on rerun with fixed seed
Training Feasibility	Trainable on single mid-range GPU

4.4 UML Diagrams

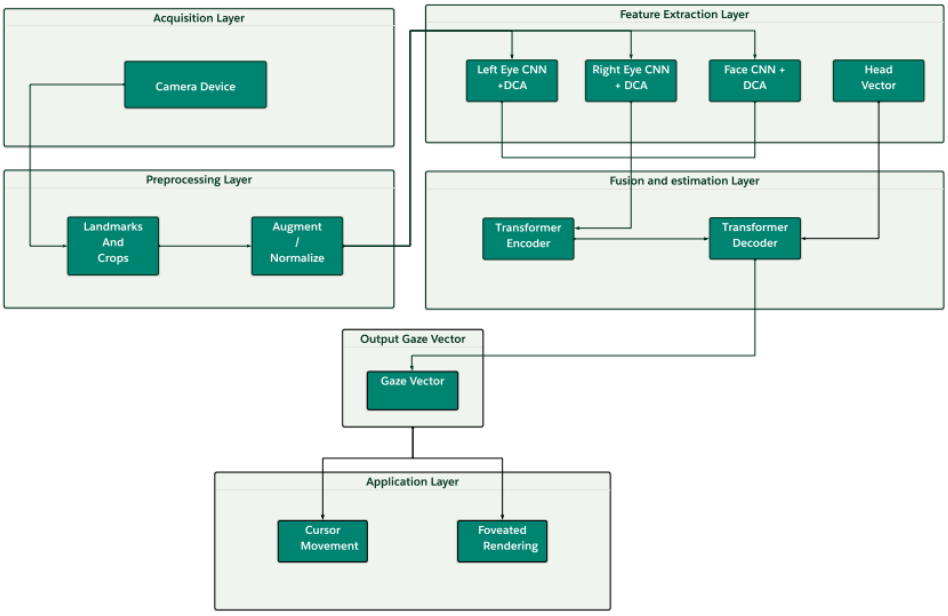

Figure 4.1: System Architecture of MetaGaze

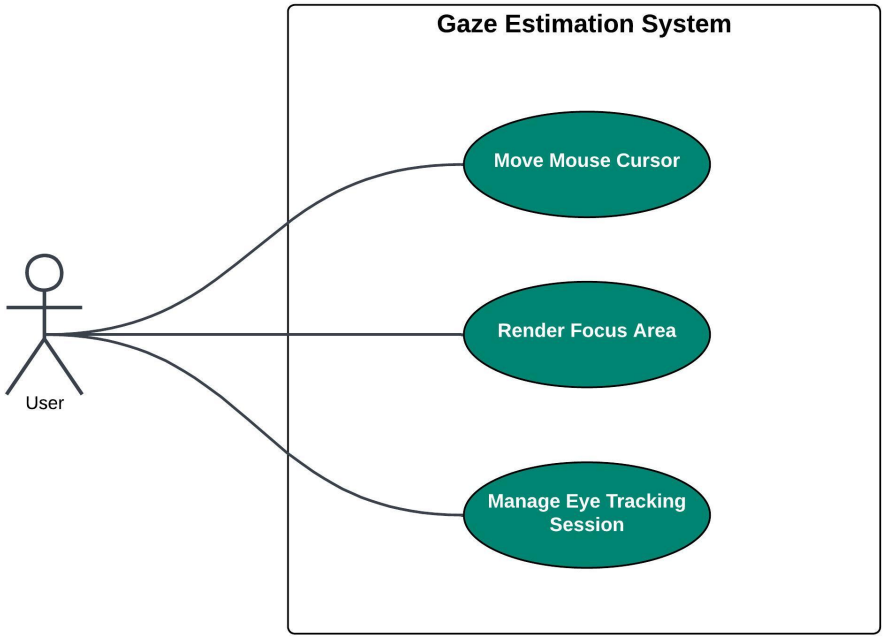

Figure 4.2: Use Case Diagram for MetaGaze

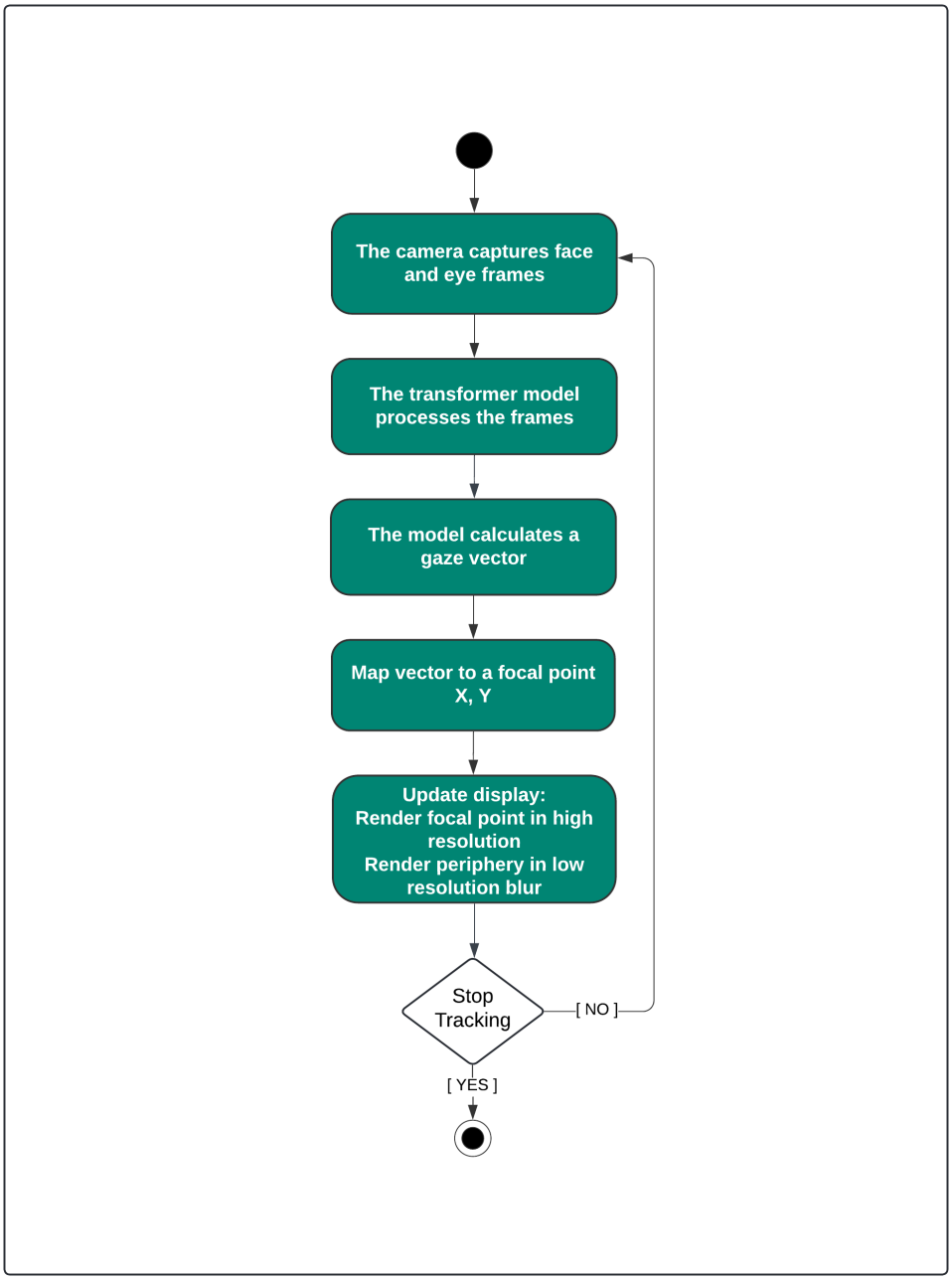

Figure 4.3: FlowChart – Mouse Cursor Movement

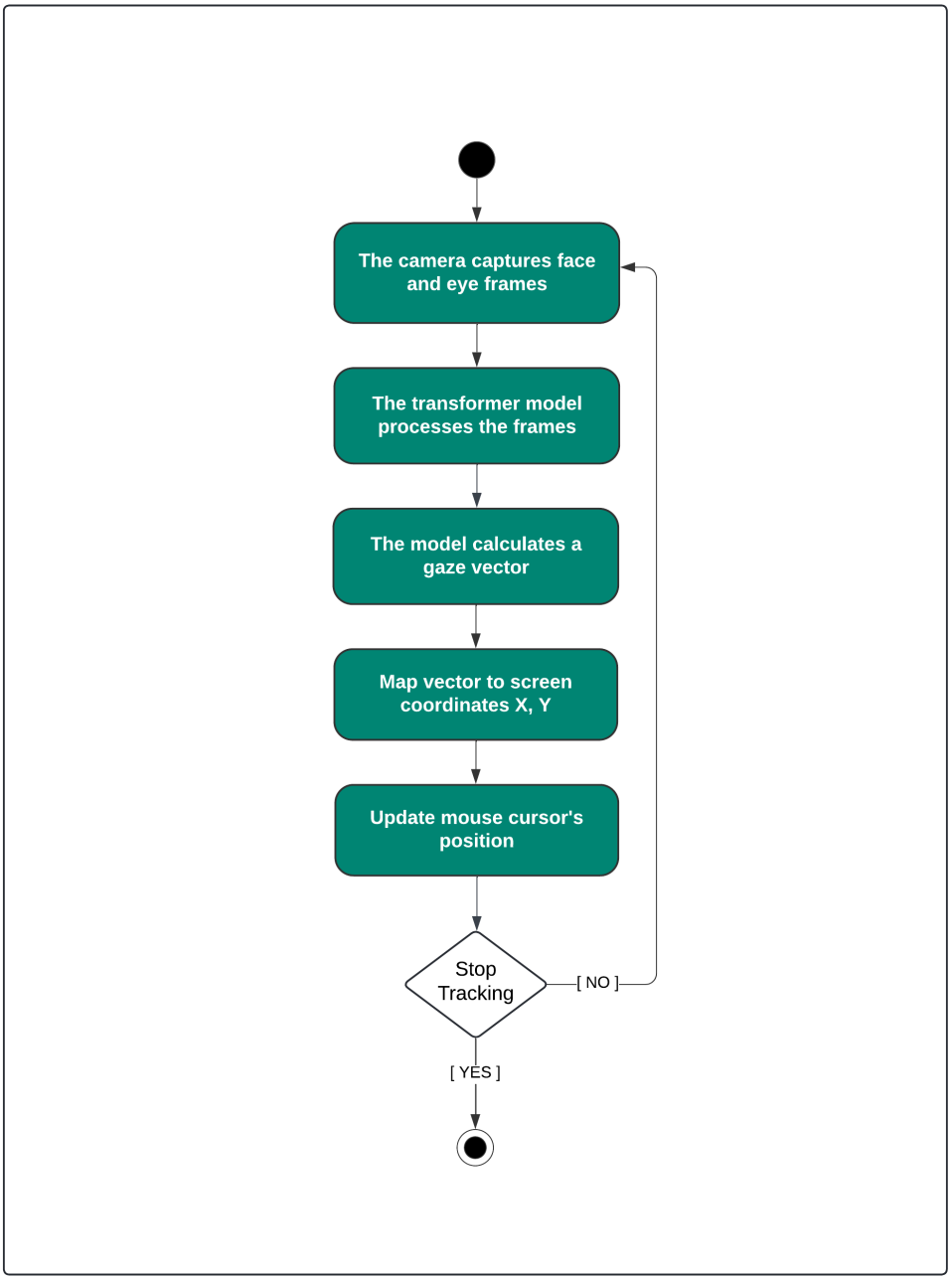

Figure 4.4: FlowChart – Foveated Rendering

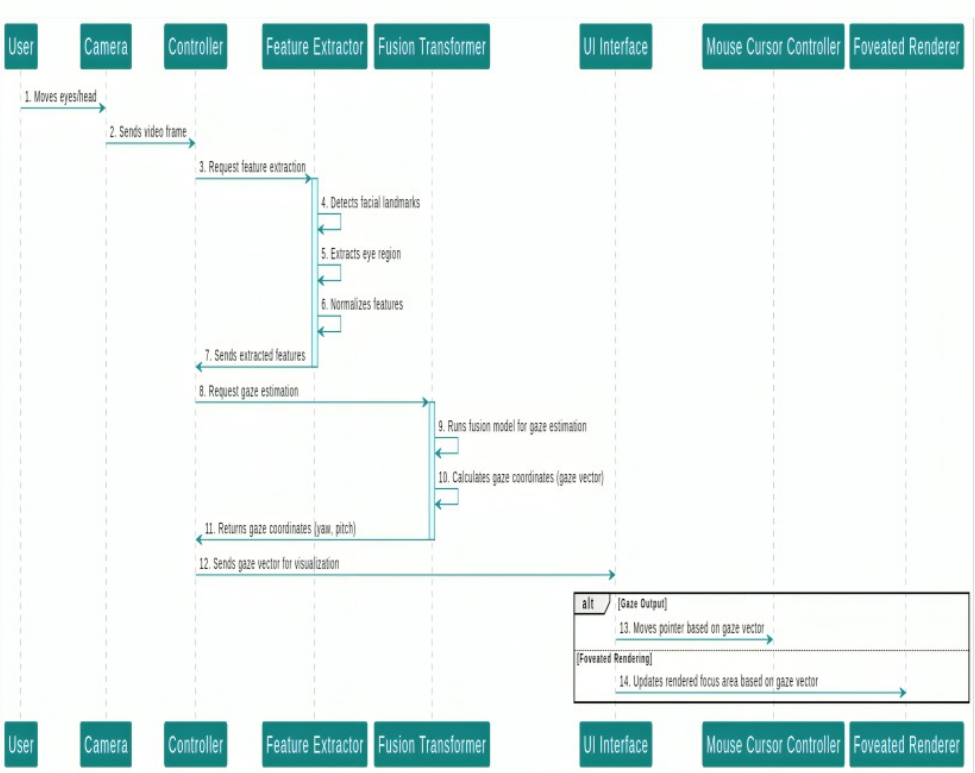

Figure 4.5: Sequence Diagram of MetaGaze Workflow

CHAPTER 5

ITERATION PLAN

5.1 Midterm FYP 1

During Iteration 1, we conducted comprehensive research aimed at understanding the landscape of gaze estimation models, specifically focusing on the transition from CNN-based methods to transformer-based architectures. We also explored essential resources and identified the ETH-XGaze dataset as the primary training source due to its diversity in head poses and lighting conditions. This phase served as a foundational step in informing subsequent project activities and architectural decisions.

5.2 Final FYP 1

In this iteration, we successfully transitioned from research to implementation. Key achievements include:

- **Preprocessing Pipeline:** Developed a pipeline to load RGB images, detect facial landmarks, and normalize eye and face crops.
- **Model Architecture:** Implemented the hybrid architecture in which CNN-extracted embeddings are fed into a transformer encoder for feature fusion.
- **Initial Training and Visualization:** Conducted preliminary training on a subset of the ETH-XGaze dataset and created a testing module that visualizes predicted gaze

vectors as 3D arrows projected onto the face.

5.3 Midterm FYP 2

The focus of the next iteration will be on scaling the training process and optimizing model performance to meet the non-functional requirements.

- **Full Dataset Training:** Expand the training loop to utilize the entire ETH-XGaze dataset (over 1 million images) to ensure robust generalization.
- **Feature Fusion Refinement:** Refine the symmetric cross-attention mechanisms within the transformer to better integrate head pose and eye features.
- **Validation:** Evaluate the model rigorously using the ETH-XGaze test set with a target angular error of approximately 3°.

5.4 Final FYP 2

The final iteration will focus on the application layer and real-time deployment, as outlined in the system architecture.

- **Real-Time Inference:** Optimize the model for real-time performance on standard webcams, eliminating the need for specialized hardware.
- **Application Integration:** Implement the downstream use cases defined in the design:
  - **Cursor Control:** Map gaze vectors to screen coordinates to control mouse movement.
  - **Foveated Rendering:** Use the gaze point to render the focus region in high resolution while blurring the periphery.

CHAPTER 6

ITERATION 1

6.1 Introduction

During Iteration 1, we conducted comprehensive research into appearance-based gaze estimation models, focusing on the transition from Convolutional Neural Networks (CNNs) to Transformer-based architectures. We explored various hybrid fusion strategies to address challenges such as head pose variations and occlusion. This foundational research step directly informed the architectural decisions for MetaGaze, particularly the choice of a hybrid CNN-Transformer framework.

6.1.1 Research Process

The research process involved a thorough exploration of academic journals and conference papers, specifically targeting methods that combine local feature extraction with global context awareness. Key activities included:

- Architectural Analysis: Investigating the limitations of pure CNNs in capturing long-range dependencies and the potential of Transformers to overcome them.
- Dataset Evaluation: We researched available public datasets to identify one that supports unconstrained environments. This led to the selection of the **ETH-XGaze** dataset due to its high diversity in head poses and lighting conditions.

6.1.2 Selected Models and Architectures

The following architectures and specific models were identified and studied to understand the state-of-the-art in gaze estimation:

- **Feature Extractors (CNNs):** ResNet architectures (e.g., ResNet-50) were studied for their efficacy in extracting fine-grained spatial features from eye and face images.
- **Pure Transformers:** Models such as *GazeTR-Pure* and *TransGaze* were analyzed to understand how self-attention mechanisms can replace standard convolutions.
- **Hybrid Fusion Models:** We focused heavily on hybrid approaches like *Res-Swin-GE* and *EM-Gaze*, which combine CNN backbones with Transformer modules to maximize robust generalization.

6.1.3 Literature Review

Detailed findings from the research conducted on the aforementioned models have been documented in the Literature Review section of Chapter 2. That section provides an in-depth analysis of each model, comparing their underlying architectures (CNN vs. Transformer), key features, and limitations regarding computational cost and environmental adaptability.

6.2 System Design

Based on the selected architectures, we formulated the high-level design for MetaGaze.

6.2.1 Use Case Diagram

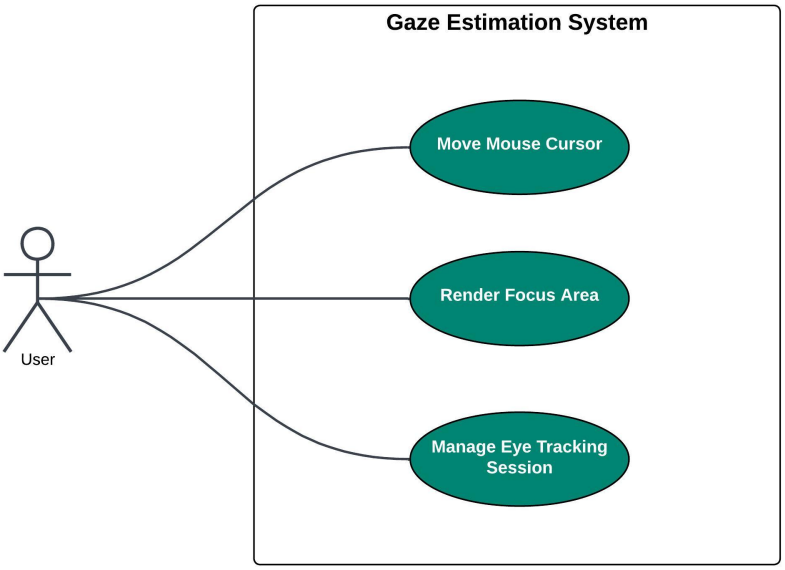

Figure 6.1: Use Case Diagram for MetaGaze

The Use Case diagram (Figure 6.1) illustrates the primary interactions between the User and the Gaze Estimation System. The system is designed to support three core functional requirements derived from our research:

- **Manage Eye Tracking Session:** The user initiates and controls the tracking session.
- **Move Mouse Cursor:** The system translates the estimated gaze vector into screen coordinates to control the cursor.
- **Render Focus Area:** The system supports foveated rendering by identifying the user's specific point of regard.

6.2.2 System Architecture

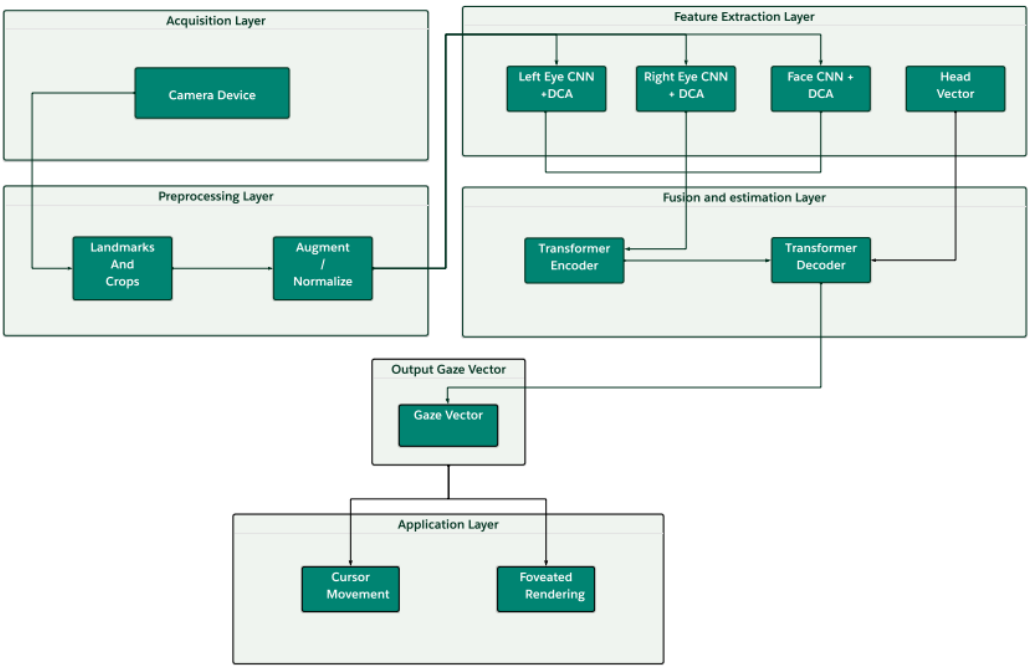

Figure 6.2: System Architecture of MetaGaze

As shown in Figure 6.2, the MetaGaze architecture is organized into distinct layers to ensure modularity and efficient processing:

- **Acquisition & Preprocessing:** Captures raw input from the camera device and processes landmarks to generate normalized crops.
- **Feature Extraction Layer:** Utilizes distinct CNN branches (Left Eye, Right Eye, and Face) to extract spatial features.
- **Fusion and Estimation Layer:** A Transformer Encoder-Decoder structure fuses these features using cross-attention mechanisms to output the final Gaze Vector.
- **Application Layer:** The output vector drives downstream applications such as cursor movement and foveated rendering.

6.3 Iteration 2: Midterm FYP 2

6.3.1 Introduction

During Iteration 2 (Midterm FYP 2), we implemented the core architectural components of MetaGaze, focusing on the Dual Cross-Attention (DCA) feature extraction mechanism and the transformer module for feature fusion. This iteration marked the transition from research and design to actual implementation, resulting in a functional basic prototype that demonstrates the system's core capabilities.

6.3.2 DCA Feature Extraction Implementation

The Dual Cross-Attention (DCA) mechanism was integrated into the ResNet backbone to enhance feature extraction capabilities. The DCA block enables the model to capture cross-modal relationships within the feature extraction process itself.

- **DCA Block Design:** Implemented a DCA module that processes three intermediate feature maps (x1, x2, x3) from the ResNet bottleneck layers. The module uses adaptive average pooling to generate global context, followed by convolutional layers and batch normalization to compute attention weights.
- **BottleneckDCA Integration:** Modified the standard ResNet Bottleneck blocks to incorporate DCA, creating BottleneckDCA blocks. This allows the attention mechanism to operate at multiple levels of the feature hierarchy, enhancing the model's ability to capture both local and global spatial relationships.
- **Multi-Stream Feature Extraction:** The ResNet50 backbone with DCA-infused bottlenecks is used to extract features from three input streams: full face, left eye, and right eye regions. Each stream processes its respective input independently, generating rich spatial feature representations.
- **Pretrained Weights:** Leveraged pretrained ResNet50 weights from ImageNet, with careful state dict matching to preserve transfer learning benefits while accommodating the DCA modifications.

6.3.3 Transformer Module Implementation

A transformer-based fusion module was implemented to integrate features from the three input modalities (face, left eye, right eye) into a unified representation for gaze prediction.

- **Feature Projection:** Implemented a linear projection layer that maps the 2048-dimensional ResNet features to a common embedding dimension (d_model=512), ensuring all modalities are represented in the same dimensional space.
- **Query Embedding:** Created learnable query embeddings that serve as the decoder input in the transformer architecture, enabling the model to learn optimal feature fusion patterns.
- **Transformer Encoder-Decoder:** Implemented a standard PyTorch Transformer with 8 attention heads, 2 encoder layers, and 2 decoder layers. The encoder processes the stacked face, left-eye, and right-eye features, while the decoder uses query embeddings to generate fused representations.
- **Attention Pooling:** Added an attention-based pooling mechanism that computes weighted combinations of the transformer output, allowing the model to focus on the most relevant features for gaze prediction.
- **Regression Head:** Implemented a multi-layer perceptron (MLP) regression head that maps the pooled features to the final gaze direction output (pitch and yaw angles).

6.3.4 Basic Prototype Development

A functional prototype was developed to demonstrate the system's capabilities and provide a user interface for testing and visualization.

- **Image Processing Pipeline:** Implemented a complete preprocessing pipeline that handles image input, face and eye region extraction using MediaPipe facial landmarks, and proper normalization for model inference.

- **Real-Time Visualization:** Added visualization capabilities that display the estimated gaze direction as an arrow overlay on the input image, providing immediate visual feedback to users.
- **Model Inference:** Integrated the complete inference pipeline, including model loading, image preprocessing, feature extraction, transformer fusion, and gaze prediction. The system outputs pitch and yaw angles in both radians and degrees.

6.3.5 Model Architecture Summary

The complete GazeSymCAT (Gaze Symmetric Cross-Attention Transformer) architecture implemented in this iteration consists of:

- **Feature Extraction:** Three ResNet50 backbones with DCA-infused Bottleneck blocks, one for each input modality (face, left eye, right eye).
- **Feature Fusion:** A transformer encoder-decoder with cross-attention mechanisms to fuse multi-modal features.
- **Gaze Prediction:** A regression head that outputs normalized gaze direction vectors (pitch, yaw).

This basic prototype successfully demonstrates the feasibility of the MetaGaze approach and provides a foundation for further enhancements in subsequent iterations.

6.4 Iteration 3: Final FYP 1

6.4.1 Introduction

Iteration 3 (Final FYP 1) focused on enhancing the MetaGaze architecture by integrating geometric head orientation vectors and implementing an improved cross-attention mechanism within the transformer architecture. This iteration represents a significant advancement, moving towards a more robust and accurate gaze estimation system that can handle varying head poses and environmental conditions.

6.4.2 Head Vector Integration

A critical enhancement in this iteration was the integration of geometric head orientation vectors into the feature fusion pipeline. The head vector provides crucial spatial context about the user's head pose (yaw, pitch, roll), which is essential for accurate gaze estimation, especially under varying head orientations.

- **Head Pose Computation:** Implemented head pose estimation using facial landmarks (MediaPipe) to compute the head orientation vector in 3D space. The system extracts rotation angles (yaw, pitch, roll) from the detected facial landmarks.
- **Feature Integration:** The head orientation vector is now properly integrated alongside facial, left-eye, and right-eye features before being fed into the transformer fusion layer. This multi-modal approach enables the model to account for head movement when estimating gaze direction.
- **Vector Normalization:** Head vectors are normalized to ensure consistent representation across different camera distances and orientations, improving the model's generalization capability.
- **Geometric Context:** The head vector provides geometric context that complements the appearance-based features extracted from images, leading to more accurate gaze predictions under challenging conditions.

6.4.3 Enhanced Cross-Attention Mechanism

The core innovation of this iteration was the refinement and proper implementation of the cross-attention mechanism within the transformer architecture. This enhanced mechanism enables the model to learn complex relationships between different feature modalities (eyes, face, and head orientation).

- **Multi-Modal Transformer Architecture:** Enhanced the Transformer Encoder-Decoder structure to handle four input modalities: face features, left-eye features, right-eye features, and head orientation vectors. Each modality is embedded into a common dimensional space.

- **Symmetric Cross-Attention:** Implemented symmetric cross-attention that allows each feature modality to attend to and integrate information from all other modalities. This bidirectional attention mechanism ensures that all feature types contribute equally to the final gaze vector prediction.
- **Attention Weight Visualization:** Added capabilities to visualize attention weights, enabling better understanding of how the model focuses on different features during gaze estimation.
- **Feature Embedding Enhancement:** Improved the feature embedding process to better accommodate the geometric head vector alongside the appearance-based CNN features, ensuring seamless integration in the transformer fusion layer.

6.4.4 Model Refinement and Optimization

In addition to the architectural improvements, this iteration included several model refinements to improve performance and robustness:

- **Training Pipeline Enhancement:** Enhanced the training pipeline to support the new multi-modal feature fusion architecture, including proper handling of head vector data during training.
- **Loss Function Optimization:** Optimized the loss function to better handle the combined features from multiple sources, ensuring balanced learning across all modalities.
- **Validation Metrics:** Implemented comprehensive validation metrics to track the improvement in gaze estimation accuracy with the integrated head vector and enhanced cross-attention mechanism.
- **Performance Optimization:** Optimized inference speed and memory usage to support real-time applications while maintaining accuracy.

6.4.5 Level 2 Design Diagrams

This iteration includes detailed Level 2 design diagrams that provide a more granular view of the system's internal structure and data flow:

6.4.5.1 Call Graph

The Call Graph diagram illustrates the function call hierarchy and dependencies within the MetaGaze system, showing how different components interact during the gaze estimation process.

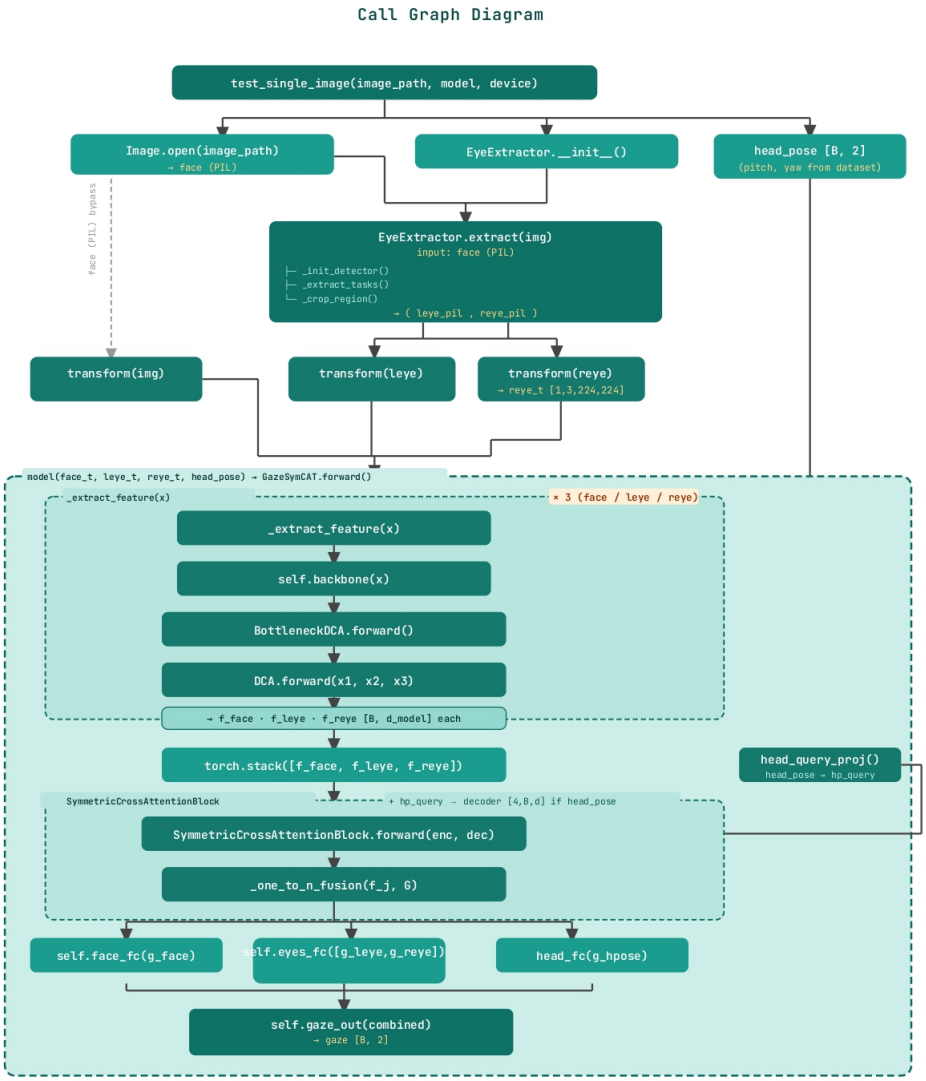

Figure 6.3: Call Graph - Function Call Hierarchy

6.4.5.2 Data Flow Diagram

The Data Flow Diagram (DFD) shows how data moves through the system, from input capture through feature extraction, fusion, and final gaze vector output.

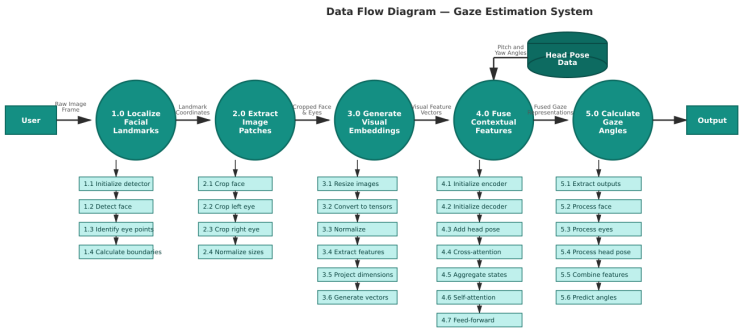

Figure 6.4: Data Flow Diagram - Level 2

6.4.5.3 Sequence Diagram

The updated Sequence Diagram (MetaGaze2) provides a detailed view of the interaction sequence between system components during a complete gaze estimation cycle, including the integration of head vector processing and cross-attention mechanisms.

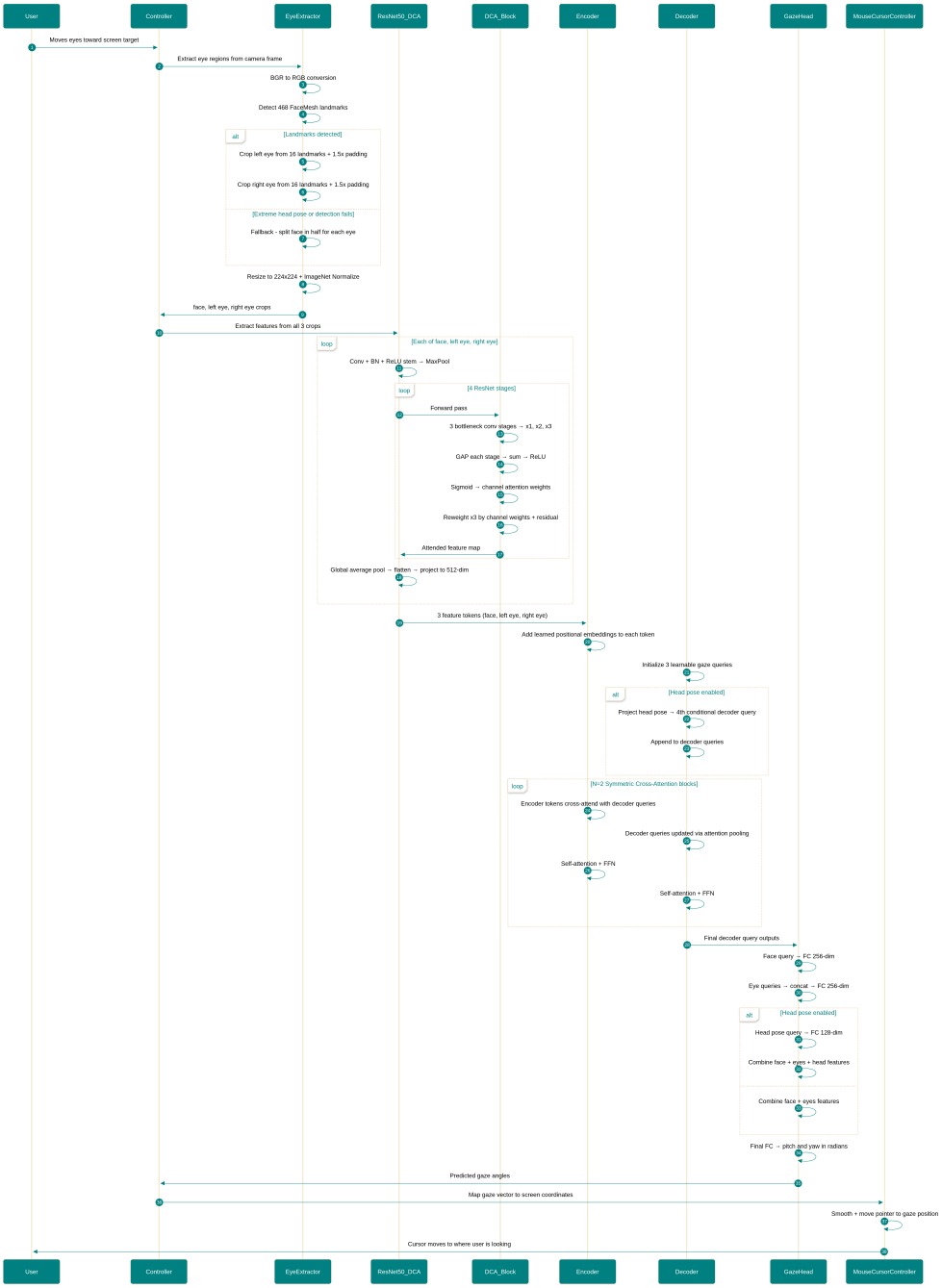

Figure 6.5: Sequence Diagram MetaGaze2 - Component Interactions

CHAPTER 7

IMPLEMENTATION DETAILS

7.1 Tools and Technologies

The MetaGaze system was implemented using a combination of modern deep learning frameworks, computer vision libraries, and development tools. This section outlines the key technologies and their roles in the implementation.

7.1.1 Deep Learning Framework

- **PyTorch:** The primary deep learning framework used for implementing the neural network architecture. PyTorch provides flexible tensor operations, automatic differentiation, and GPU acceleration support, making it ideal for research and development of the MetaGaze model.

- **Torchvision:** Used for accessing pretrained ResNet models and standard image transformations. The ResNet50 architecture with pretrained ImageNet weights serves as the backbone for feature extraction.

- **NumPy:** Essential for numerical computations, array operations, and data preprocessing throughout the pipeline.

7.1.2 Computer Vision Libraries

- **MediaPipe:** Utilized for facial landmark detection and head pose estimation. MediaPipe provides robust, real-time face detection and landmark extraction capabilities that are essential for preprocessing input images.
- **OpenCV:** Used for image processing operations, including image resizing, cropping, normalization, and visualization of gaze vectors on images.
- **PIL (Pillow):** Python Imaging Library employed for image loading, format conversion, and basic image manipulation tasks.

7.1.3 Development Environment

- **Python 3.8+:** The primary programming language for the entire implementation.
- **CUDA:** GPU acceleration support for training and inference, significantly reducing computation time.
- **Git:** Version control system for collaborative development and code management.

7.2 Model Architecture Implementation

The MetaGaze architecture consists of several key components, each implemented as modular Python classes for maintainability and extensibility.

7.2.1 DCA Block Implementation

The Dual Cross-Attention (DCA) mechanism is implemented as a custom PyTorch module that processes three intermediate feature maps from ResNet bottleneck layers. The DCA block computes attention weights by:

- Applying adaptive average pooling to generate global context from each feature map
- Processing pooled features through convolutional layers with batch normalization

- Computing attention weights using a sigmoid activation function
- Applying the attention weights to the final feature map

7.2.2 ResNet Backbone with DCA

The standard ResNet50 architecture is modified to incorporate DCA blocks within each Bottleneck layer. This is achieved by:

- Creating a custom `BottleneckDCA` class that inherits from PyTorch's `Bottleneck` class
- Integrating the DCA module to process intermediate feature maps (x1, x2, x3) within each bottleneck
- Loading pretrained ImageNet weights while accommodating the DCA modifications

7.2.3 Transformer Fusion Module

The transformer-based feature fusion is implemented using PyTorch's native `nn.Transformer` module:

- **Feature Projection:** A linear layer maps 2048-dimensional ResNet features to a common embedding dimension (d_model=512)
- **Query Embeddings:** Learnable parameter tensors serve as decoder queries for the transformer
- **Multi-Head Attention:** The transformer uses 8 attention heads with 2 encoder and 2 decoder layers
- **Attention Pooling:** A custom pooling mechanism computes weighted combinations of transformer outputs
- **Regression Head:** A multi-layer perceptron (MLP) with dropout maps pooled features to gaze angles

7.3 Data Preprocessing Pipeline

The preprocessing pipeline handles input images and prepares them for model inference.

7.3.1 Facial Landmark Detection

- MediaPipe Face Mesh is used to detect 468 facial landmarks
- Landmarks are used to extract normalized eye regions (left and right) and full face regions
- Eye regions are cropped and resized to 224×224 pixels for consistent input dimensions

7.3.2 Image Normalization

- Images are normalized using ImageNet statistics (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225])
- Normalization ensures compatibility with pretrained ResNet weights
- Images are converted to PyTorch tensors with proper channel ordering (RGB)

7.4 Training Infrastructure

The training pipeline is designed to handle large-scale datasets like ETH-XGaze efficiently.

7.4.1 Dataset Handling

- Custom PyTorch `Dataset` class for loading and preprocessing ETH-XGaze images
- Efficient data loading with multiple worker processes
- Data augmentation techniques (random cropping, flipping) for improved generalization

7.4.2 Training Configuration

- **Optimizer:** Adam optimizer with learning rate scheduling
- **Loss Function:** Mean Squared Error (MSE) or L1 loss for gaze angle regression
- **Batch Size:** Configured based on available GPU memory
- **Validation:** Regular validation on held-out test sets to monitor training progress

7.5 Code Organization

The codebase is organized into modular components for maintainability:

- **model.py:** Contains the main MetaGaze architecture and DCA blocks
- **dataset.py:** Dataset loading and preprocessing utilities
- **utils.py:** Helper functions for visualization, image processing, and evaluation
- **config.py:** Configuration parameters and hyperparameter settings
- **train.py:** Training script with logging and checkpointing
- **evaluate.py:** Evaluation script for computing metrics on test sets

This modular structure facilitates code reuse, testing, and future extensions to the MetaGaze system.

APPENDIX A

APPENDIX I TITLE

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